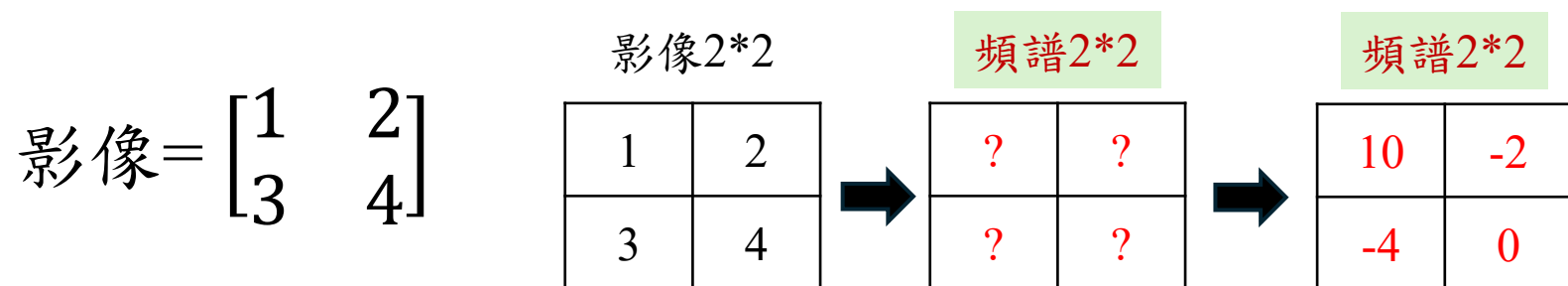


Example 1: 傅立葉變換

- 要將一張寬×高=2×2的影像進行傅立葉轉換，我們可以使用離散傅立葉變換（DFT）公式來計算其頻譜。求傅立葉變換後的 $F(0,0)$



會使用到的公式：

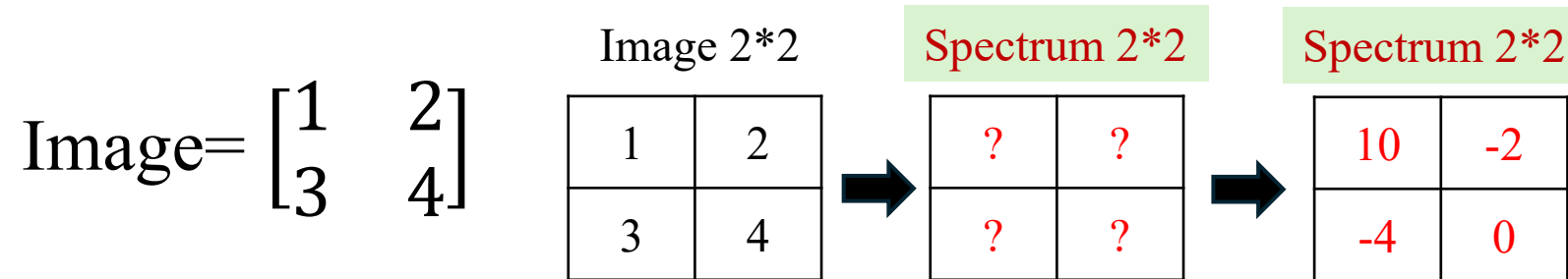
$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(0,0) = f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 1}{2}\right)}$$

$$F(0,0) = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 = 10$$

Example 1: Fourier Transform

- To perform a Fourier Transform on a 2x2 image, we can use the Discrete Fourier Transform (DFT) formula to compute its spectrum. Find the value of $F(0,0)$ after the Fourier Transform.



The formula to be used:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(0,0) = f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 1}{2}\right)}$$

$$F(0,0) = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 = 10$$

Example 2: Fourier Transform

頻譜2*2

1	2
3	4



Spectrum 2*2

	F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Image 2*2	1	2	Spectrum 2*2	10	
	3	4			F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(1,1) = f(0,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)}$$

$$e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} = e^{-j2\pi\left(\frac{2}{2}\right)} = e^{-j2\pi} \quad \text{Euler's formula!} \quad e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$(\theta = -2\pi)$$

$$e^{-j2\pi} = \cos(-2\pi) + j \sin(-2\pi) = 1 + j0 = 1$$

$$e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} = e^{-j2\pi\left(\frac{1}{2}\right)} = e^{-j\pi}$$

$$(\theta = -\pi)$$

$$e^{-j\pi} = \cos(-\pi) + j \sin(-\pi) = -1 + j0 = -1$$

Image 2*2	1	2
	3	4

Spectrum 2*2	10	
		F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(1,1) = f(0,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)}$$

$$e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} = e^{-j2\pi\left(\frac{1}{2}\right)} = e^{-j\pi}$$

Euler's formula!

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$(\theta = -\pi)$$

$$e^{-j\pi} = \cos(-\pi) + j \sin(-\pi) = -1 + j0 = -1$$

$$e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)} = e^{-j2\pi\left(\frac{2}{2}\right)} = e^{-j2\pi} = \cos(-2\pi) + j \sin(-2\pi) = 1 + j0 = 1$$

$$F(1,1) = 1 \cdot (1) + 2 \cdot (-1) + 3 \cdot (-1) + 4 \cdot (1) = 0$$

頻譜2*2

10	?
?	0

Example 3 : Fourier Transform

Image $f(x,y)$ 3*3

1	2	3
4	5	6
7	8	9

Spectrum $|F(u,v)|$ 3*3

$ F(0,1) $		

Q: Find the magnitude of $|F(0,1)|$?

Fourier Transform

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Magnitude Spectrum

$$Magnitude = |F(u, v)| = \sqrt{\text{實部}(F(u, v))^2 + \text{虛部}(F(u, v))^2}$$

$$|F(u, v)| = \sqrt{(Re(F(u, v)))^2 + (Im(F(u, v)))^2}$$

Fourier Spectrum

Fourier Transform

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

1 radian = $180/\pi$ degrees

$$\begin{aligned}
 F(0,1) &= f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 0}{3}\right)} & + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 0}{3}\right)} & + f(2,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 0}{3}\right)} \\
 &+ f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 1}{3}\right)} & + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 1}{3}\right)} & + f(2,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 1}{3}\right)} \\
 &+ f(0,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 2}{3}\right)} & + f(1,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 2}{3}\right)} & + f(2,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 2}{3}\right)}
 \end{aligned}$$



$$\begin{aligned}
 F(0,1) &= 1 \cdot e^{-j\left(\frac{0\pi}{3}\right)} & + 2 \cdot e^{-j\left(\frac{0\pi}{3}\right)} & + 3 \cdot e^{-j\left(\frac{0\pi}{3}\right)} \\
 &+ 4 \cdot e^{-j\left(\frac{2\pi}{3}\right)} & + 5 \cdot e^{-j\left(\frac{2\pi}{3}\right)} & + 6 \cdot e^{-j\left(\frac{2\pi}{3}\right)} \\
 &+ 7 \cdot e^{-j\left(\frac{4\pi}{3}\right)} & + 8 \cdot e^{-j\left(\frac{4\pi}{3}\right)} & + 9 \cdot e^{-j\left(\frac{4\pi}{3}\right)}
 \end{aligned}$$

$$\frac{4 \cdot 180}{3} = 240^\circ$$

Fourier Spectrum

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$\begin{aligned} F(0,1) = & 1 * (\cos(0^\circ) - j\sin(0^\circ)) & +2 * (\cos(0^\circ) - j\sin(0^\circ)) & +3 * (\cos(0^\circ) - j\sin(0^\circ)) \\ & + 4 * (\cos(120^\circ) - j\sin(120^\circ)) & +5 * (\cos(120^\circ) - j\sin(120^\circ)) & +6 * (\cos(120^\circ) - j\sin(120^\circ)) \\ & + 7 * (\cos(240^\circ) - j\sin(240^\circ)) & +8 * (\cos(240^\circ) - j\sin(240^\circ)) & +9 * (\cos(120^\circ) - j\sin(240^\circ)) \end{aligned}$$



$$\begin{aligned} F(0,1) = & 1 * (1 - j0) & +2 * (1 - j0) & +3 * (1 - j0) \\ & + 4 * (-0.5 - j0.866) & +5 * (-0.5 - j0.866) & +6 * (-0.5 - j0.866) \\ & + 7 * (-0.5 - j(-0.866)) & +8 * (-0.5 - j(-0.866)) & +9 * (-0.5 - j(-0.866)) \end{aligned}$$

Fourier Spectrum

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$F(0,1) = (1 - 0j)$$

$$+(-2 - j3.464)$$

$$+(-3.5 + j(6.062))$$

$$+(2 - j0)$$

$$+(-2.5 - j4.33)$$

$$+(-4 + j6.928)$$

$$+(3 - j0)$$

$$+(-3 - j5.196)$$

$$+(-4.5 + j7.794)$$

$$F(0,1) = -13.5 + 7.794j$$

$$Magnitude = |F(u, v)| = \sqrt{\text{實部}(F(u, v))^2 + \text{虛部}(F(u, v))^2}$$

$$|F(0,1)| = \sqrt{(-13.5)^2 + (7.794)^2} = 15.588 \#$$

Spectrum $|F(u, v)|$

15.588		

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